

Chi2 interpretation

Chi squared is the output of the fit parameters, and is "how many variances away" is each data point.

PTE: the probability that the χ^2 would be bigger than the one we got. We hope for PTE (aka $P_{>}$) = 0.5.

$$\chi^2 = \sum_{m=1}^M \left(\frac{\bar{y}_m - f^{fit}(x_m)}{S_{\bar{y}_m}} \right)^2 \quad (1)$$

$$\left\langle \frac{\chi^2}{M - p} \right\rangle = 1 \quad (2)$$

$$(3)$$

Where M is the number of data points across the range (not multiple measurements of one data point), and p is the number of degrees of freedom.

We subtract p , because we can always force a 0 for p number of data points in the fit (i.e. a P^3 can be constructed to hit exactly where three points land). There are $M - p$ data points that aren't guaranteed. Reducing the χ^2 actually smears the error around all data points, so there are no 0 but the others are also less than 1.

If $\chi^2 \ll 1$

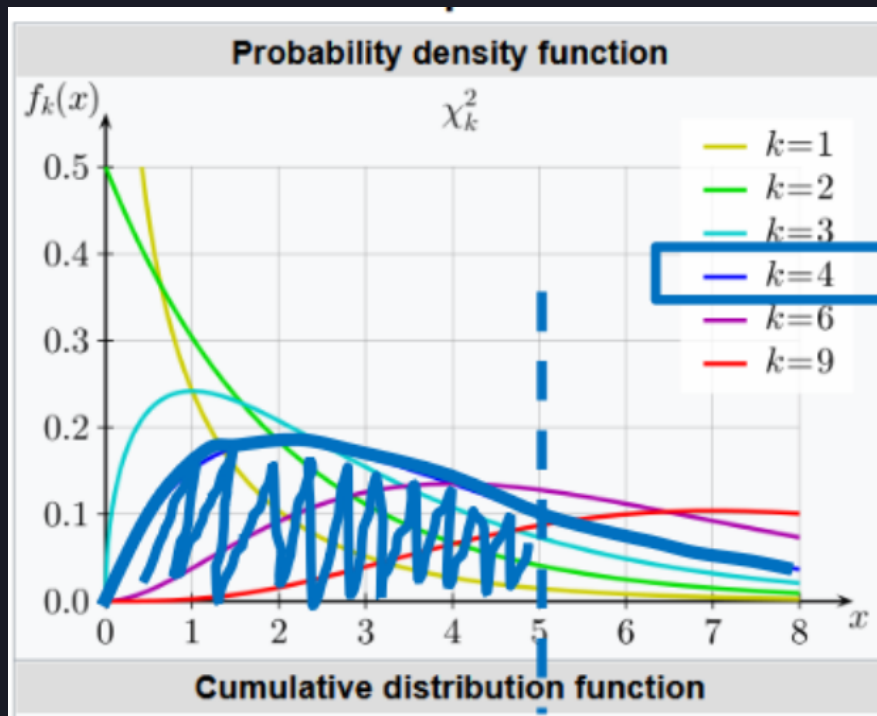
Either really lucky, error bars are way too large, or the errors across each data point in a series (not all one measurement point) are correlated. Errors not being random biases one way or another.

if $\chi^2 \gg 1$

Either a really unlucky, really bad fit, or error bars are way too small/.

How do we know when we were lucky/unlucky, or when we need to check experiment / fit?

- Number of degrees of freedom relative to number of data points
- χ^2 is the sum of the squares of $M - p$ standard normal random variables (mean 0, variance 1 because we subtracted the mean, and divide by the variance).
- PTE is a guide:



- We have a probability distribution for what χ^2 we got given the number of degrees of freedom. $(M - k)$ where k is the number of data points.
- What is the probability that someone would get something bigger than my χ^2 because of random error.
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